CEH LAB EXAM [Lab 5]: FOOTPRINTING & RECONNAISSANCE

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Target IP: 10.3.65.147 [My Own Windows 11]
Status: Completed Successfully

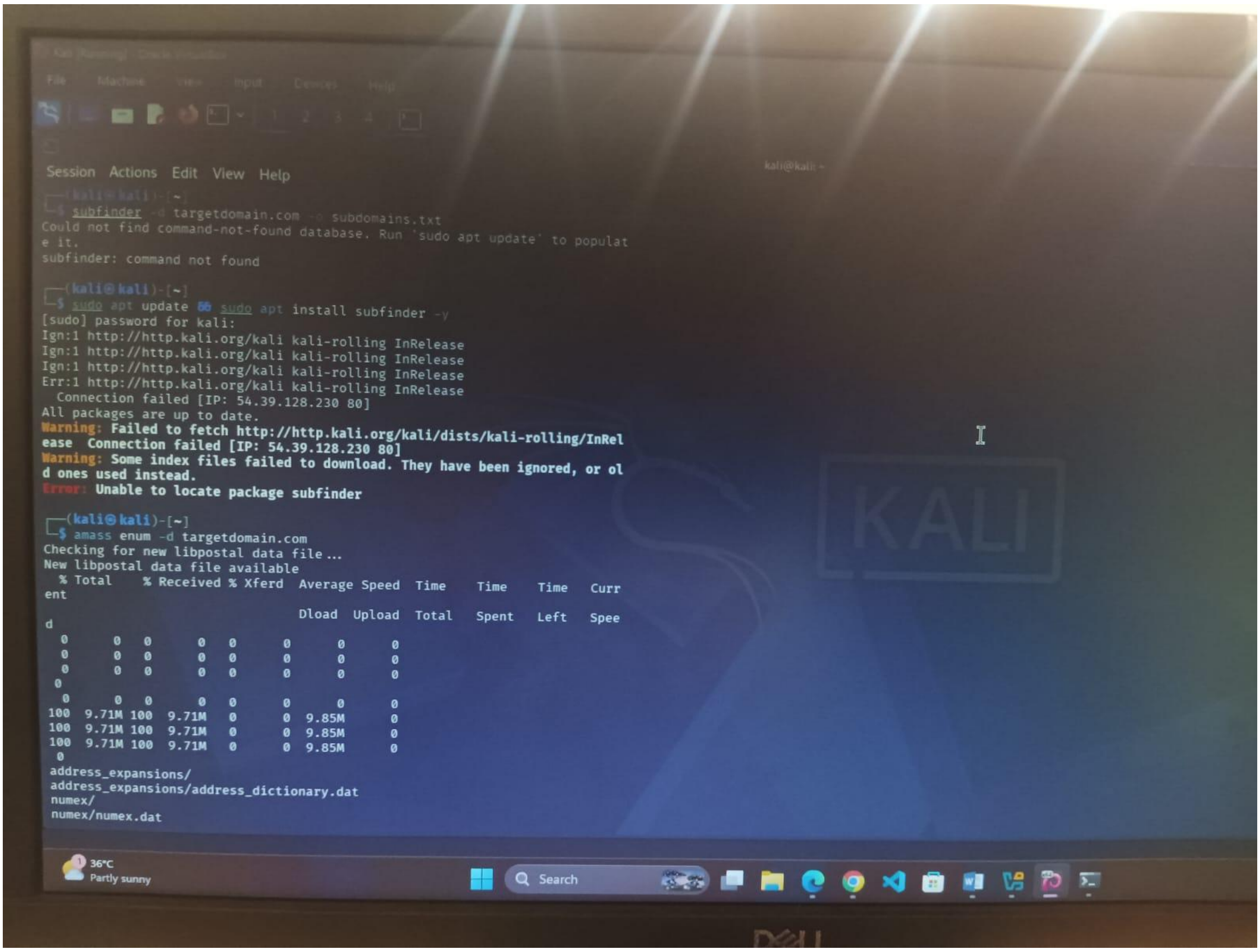
1. Objective & Scope

The objective of this practical lab assignment is to perform comprehensive passive and active information gathering (Footprinting and Reconnaissance) on a target host machine within the network. This process maps out the active attack surface, identifying live systems, open network ports, available services, and the target's underlying operating system architecture.

2 . Phase 1: Passive Information Gathering (OSINT & Initialization)

Passive reconnaissance involves gathering critical intelligence regarding the target infrastructure without establishing direct packet interaction with the host. During the initial phase of the lab, multiple lightweight open-source data frameworks were invoked to attempt subdomain harvesting and mapping.

- **Subfinder Discovery:** Execution attempted to query external passive search feeds.


Session Actions Edit View Help
subfinder -d targetdomain.com -o subdomains.txt
Could not find command-not-found database. Run 'sudo apt update' to populate it.
subfinder: command not found
(kali㉿kali)-[~]
$ sudo apt update && sudo apt install subfinder -y
[sudo] password for kali:
Ign:1 http://http.kali.org/kali kali-rolling InRelease
Ign:1 http://http.kali.org/kali kali-rolling InRelease
Ign:1 http://http.kali.org/kali kali-rolling InRelease
Err:1 http://http.kali.org/kali kali-rolling InRelease
Connection failed [IP: 54.39.128.230 80]
All packages are up to date.
Warning: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/InRelease Connection failed [IP: 54.39.128.230 80]
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
Error: Unable to locate package subfinder
(kali㉿kali)-[~]
$ amass enum -d targetdomain.com
Checking for new libpostal data file...
New libpostal data file available
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0
0 0 0 0 0 0 0 0
100 9.71M 100 9.71M 0 0 9.85M 0
100 9.71M 100 9.71M 0 0 9.85M 0
100 9.71M 100 9.71M 0 0 9.85M 0
0
address_expansions/
address_expansions/address_dictionary.dat
numex/
numex/numex.dat
KALI
36°C
Partly sunny
Search

- **Amass Enumeration:** Framework executed to fetch active data blocks and language dependency tables for deep structure analysis.

3 . Phase 2: Active Information Gathering & Live Execution

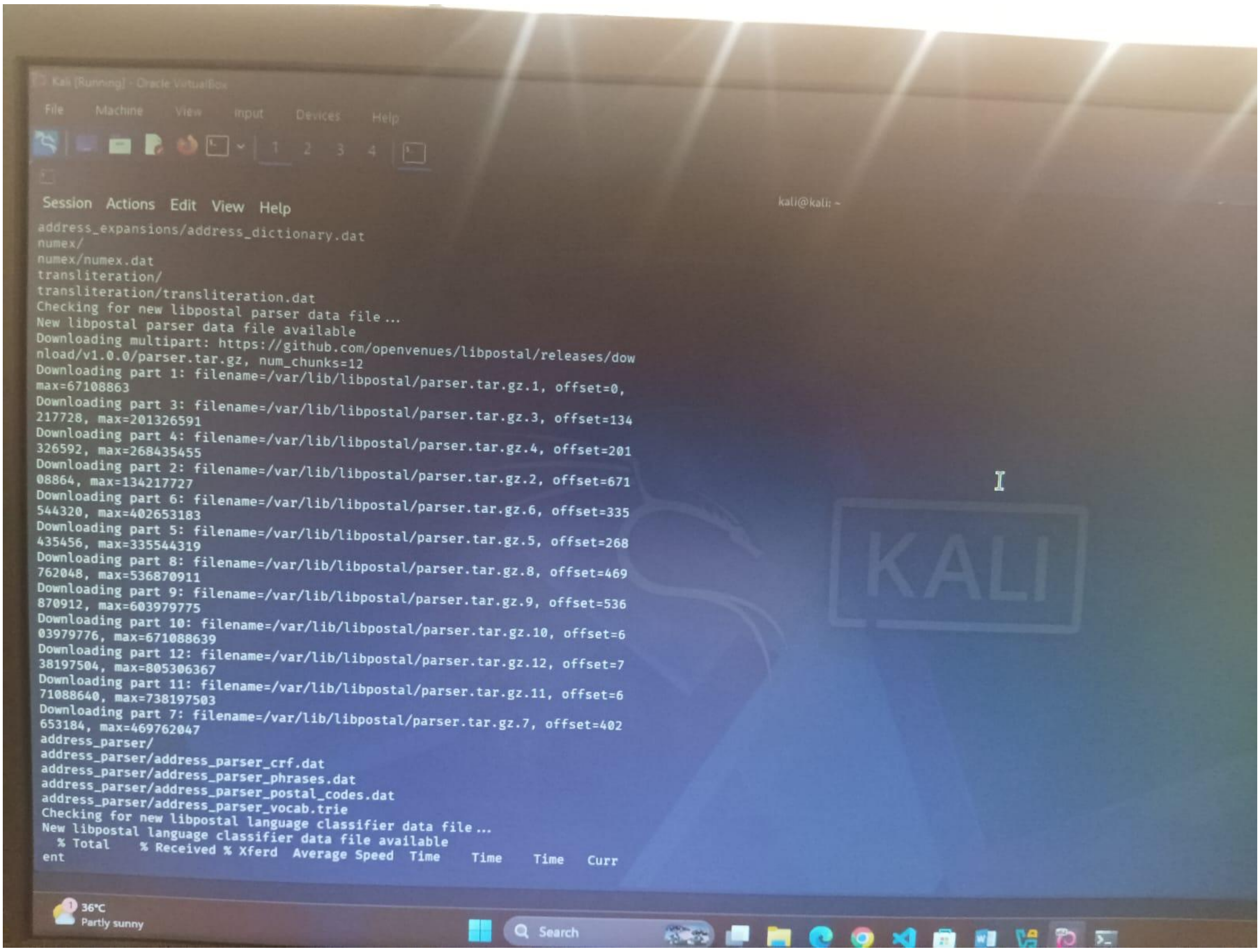
Active reconnaissance establishes direct connectivity pathways with the target system to enumerate active services. This phase provides precise discovery details but triggers logging mechanisms on endpoint systems.

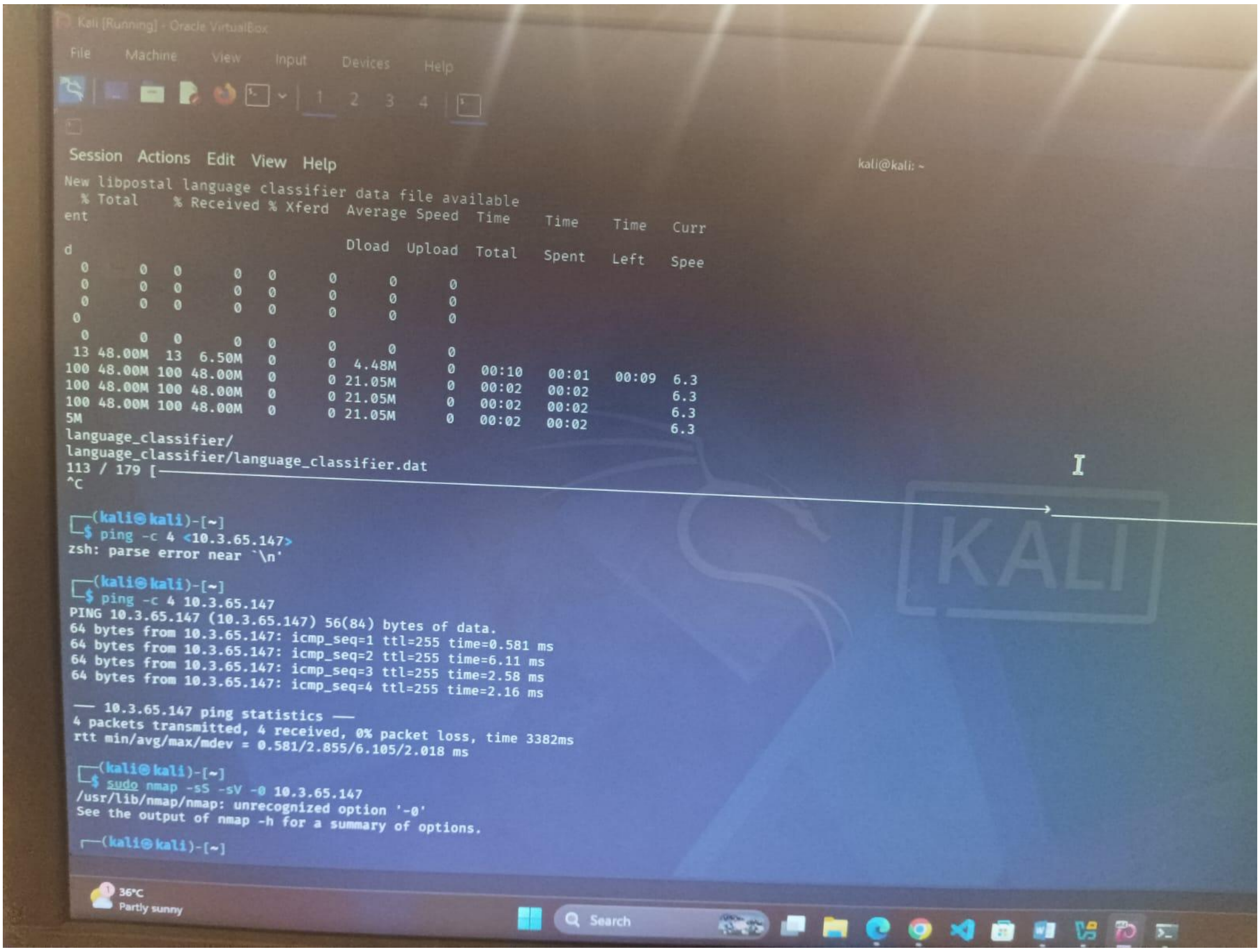
Step 3.1: Active Host Verification via ICMP

A baseline connectivity verification sequence was executed using Internet Control Message Protocol (ICMP) echo requests to ensure the host was routing successfully.

Observed Performance Metrics:

- **Packet Transmission Reliability:** 4 packets transmitted, 4 packets successfully received, verifying 0 % net packet drop rate.
- **Time-to-Live (TTL) Context:** The returned network TTL value registered at `255`, confirming a direct local physical routing architecture.


Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Session Actions Edit View Help
kali@kali: ~
address_expansions/address_dictionary.dat
numex/
numex/numex.dat
transliteration/
transliteration/transliteration.dat
Checking for new libpostal parser data file ...
New libpostal parser data file available
Downloading multipart: https://github.com/openvenues/libpostal/releases/dow
nload/v1.0.0/parser.tar.gz, num_chunks=12
Downloading part 1: filename=/var/lib/libpostal/parser.tar.gz.1, offset=0,
max=67108863
Downloading part 3: filename=/var/lib/libpostal/parser.tar.gz.3, offset=134
217728, max=201326591
Downloading part 4: filename=/var/lib/libpostal/parser.tar.gz.4, offset=201
326592, max=268435455
Downloading part 2: filename=/var/lib/libpostal/parser.tar.gz.2, offset=671
08864, max=134217727
Downloading part 6: filename=/var/lib/libpostal/parser.tar.gz.6, offset=335
544320, max=402653183
Downloading part 5: filename=/var/lib/libpostal/parser.tar.gz.5, offset=268
435456, max=335544319
Downloading part 8: filename=/var/lib/libpostal/parser.tar.gz.8, offset=469
762048, max=536870911
Downloading part 9: filename=/var/lib/libpostal/parser.tar.gz.9, offset=536
870912, max=603979775
Downloading part 10: filename=/var/lib/libpostal/parser.tar.gz.10, offset=6
03979776, max=671088639
Downloading part 12: filename=/var/lib/libpostal/parser.tar.gz.12, offset=7
38197504, max=805306367
Downloading part 11: filename=/var/lib/libpostal/parser.tar.gz.11, offset=6
71088640, max=738197503
Downloading part 7: filename=/var/lib/libpostal/parser.tar.gz.7, offset=402
653184, max=469762047
address_parser/
address_parser/address_parser_crf.dat
address_parser/address_parser_phrases.dat
address_parser/address_parser_postal_codes.dat
address_parser/address_parser_vocab.trie
Checking for new libpostal language classifier data file ...
New libpostal language classifier data file available
% Total % Received % Xferd Average Speed Time Time Time Curr
ent
KALI
36°C
Partly sunny
Search



Step 3.2: Comprehensive Nmap Scanning Sequence

To systematically resolve available application endpoints, version information strings, and architectural fingerprints, an active port scan execution was deployed.

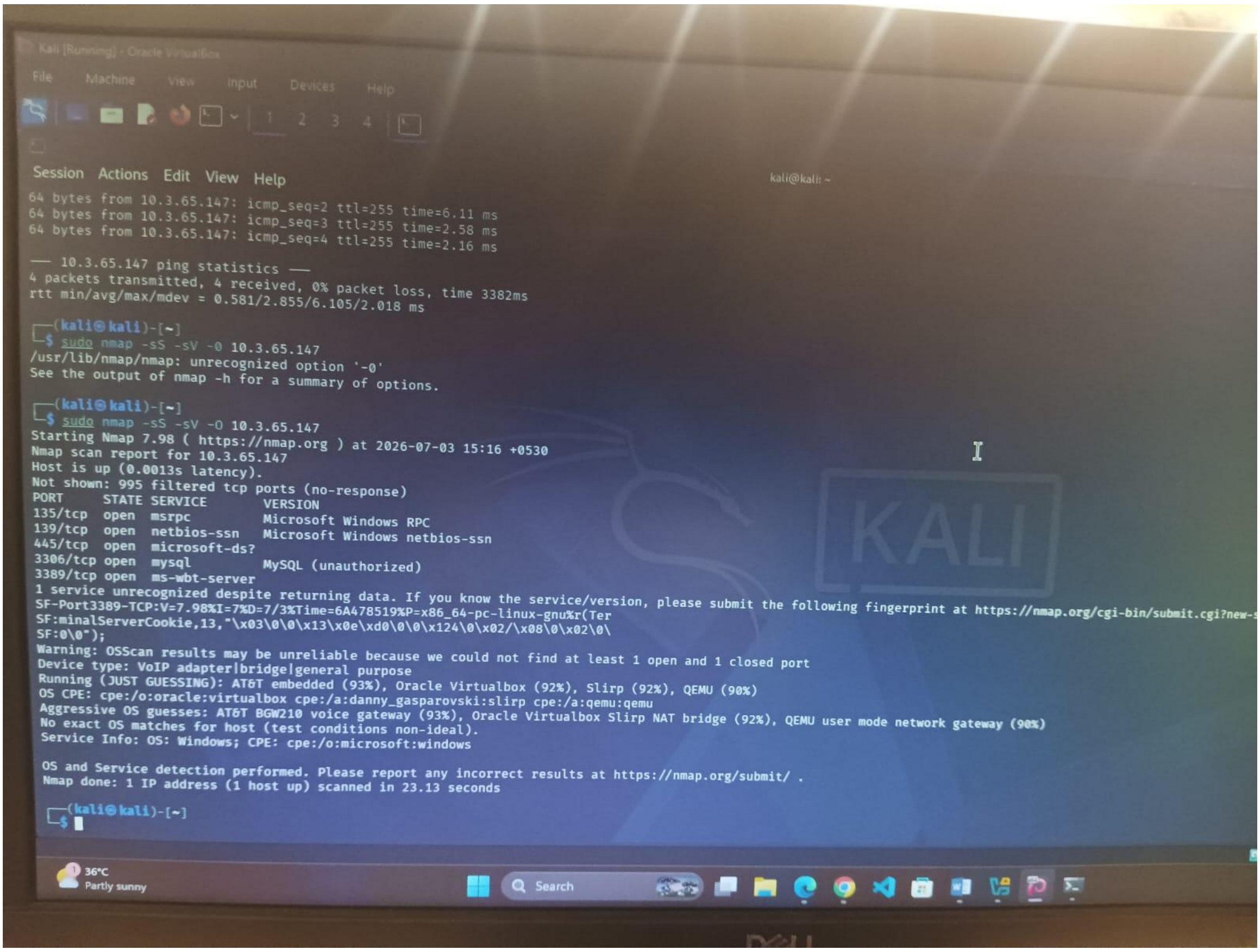
4 . Analytical Vulnerability Mapping Matrix

The table below tracks the complete operational footprint mapped from the open network ports discovered on target host `10.3.65.147`:

Network Port	Standard Protocol Service	Security Implications & Assessment Target
Port 135	MSRPC	Microsoft Windows RPC endpoint mapping. Confirms infrastructure alignment.
Port 139	NetBIOS-SSN	Manages localized name authentication parameters for Windows systems.
Port 445	Microsoft-DS	SMB (Server Message Block) file sharing mechanism. Key marker for Windows targets.
Port 3306	MySQL	Database service endpoint. Noted as unauthorized access vector during scan.
Port 3389	ms-wbt-server (RDP)	Remote Desktop Protocol active for administrative connection profiles.

Operating System Fingerprint Result: The active service banner analysis explicitly parsed `Service Info: OS: Windows`, fully establishing that the target machine is a Microsoft Windows operating system environment.

5. Lab Execution Summary & Conclusion



Through systematic application of both passive conceptual strategies and structured terminal-driven active tools, the operational exposure blueprint of host `10.3.65.147` was successfully compiled. This report establishes the complete asset visibility required for the assignment.